Small Edges, Strict Boundaries: A Chronological Evaluation of Intraday Stock Direction Classification

Anonymous Author(s)

Abstract

In intraday 5-minute stock-direction classification, validation contact, leakage, and repeated comparison can inflate macro-F1 gains. We contribute, not a new model: a strict pre-registered protocol; under it, a small, sign-consistent TCN margin over same-row baselines; and calibration and activity-regime diagnostics that locate where it holds and breaks. Before scoring, we freeze the label, no-trade band, chronological splits, roster, and decision rules. Each model is compared with a same-row stratified dummy, and each validation look is logged in a counted validation-budget ledger. On the frozen validation split, a predeclared TCN exceeds that dummy by 1.69 pp in macro-F1 (two-seed mean, 1.63 pp at the worse seed; too few to quantify variance). A separate guarded walk-forward over the post-2017 segment, historically contacted and not a clean test, stays positive (row-pooled +0.636 pp) and clears a deliberately weak predeclared bar. A descriptive CSCV/PBO readout of 0.514, near a coin flip, is too coarse to rule overfitting in or out. The residual sits in the calmest bars and drops below the balanced random prior on the most active bars in both windows (one historically contacted), a pattern we cannot separate from a bid-ask-bounce artifact. We measure macro-F1, not economic value. The protocol is shown on five large-cap US equities as one near-null case with no known-signal positive control, so its sensitivity to a genuine signal is untested.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Mathematics of computing** → *Probability and statistics*; • **Applied computing** → *Economics.*

Keywords

intraday direction classification, chronological validation, no-trade band, model calibration, selective classification, backtest overfitting

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1 Introduction

A 1.69 pp macro-F1 margin can be a signal, or it can be an artifact of how a weak intraday classification task was evaluated. The target sits near a same-row stratified dummy floor, so validation contact, temporal leakage, and repeated model comparison can change the apparent conclusion. Financial machine-learning studies have applied flexible models to stock and limit-order book prediction [15, 39]. A separate literature shows why adaptive validation, data-snooping, and backtest overfitting can inflate empirical gains [3, 17, 22]. This paper asks a narrower question: before any trading backtest, what remains when a return-free, drop-neutral direction-classification estimand is scored with frozen labels, chronological day-block splits, a fixed roster, and a counted validation budget?

The contribution is the evaluation discipline, not a new architecture, and it has three parts. First, a strict pre-registered protocol freezes the label and band, split, window, roster, same-row dummy floor, and validation-budget ledger before scoring. Second, under that protocol the predeclared TCN primary exceeds the same-row stratified dummy and the majority floor by a small margin that is sign-consistent when pooled but not uniform across tickers, periods, or regimes; we report it and select no model. Third, calibration, selective-prediction, and activity-tercile diagnostics locate where that margin concentrates and where it breaks down – a map of the edge, not a trading result. The guarded historically-contacted walk-forward applies the same discipline past the validation boundary; it is protocol evidence, not a separate headline contribution.

The worked case is deliberately narrow. It covers five large-cap US survivors, 2 training seeds, and macro-F1 only; it has no P&L, cost model, Sharpe ratio, or known-signal positive control. On the frozen validation split, the predeclared TCN primary reaches macro-F1 of 0.5170 ± 0.0009 and exceeds the same-row stratified dummy by 1.69 pp in the seed mean (row-pooled). Separately, in the train-inner control domain, four control rows, two of them last-step degenerate controls, span only 0.66 pp in macro-F1. That spread is not a full-family validation comparison; it is context for the thin validation-domain margin.

The guarded readout keeps the same restraint. The predeclared TCN primary meets the guarded stability bar, but the bar is intentionally weak: at least two positive periods out of seven and a positive pooled delta. The observed 5/7 positive periods are descriptive. The binding guarded row-pooled delta is +0.636 pp, with an equal-weight companion of +0.550 pp. A descriptive row-pooled multiplicity check gives a PBO readout of 0.514, and only the predeclared primary's period-LCB clears zero; no family is selected.

The diagnostics narrow the claim further. Low calibration error (≈ 0.010 equal-mass ECE) appears with near-zero Brier resolution, so confidence is not useful enough to define an operating rule. The activity-tercile map is also a boundary. In validation, the low-activity slice is +5.43 pp over dummy while the high-activity slice is -1.54 pp; the same low-to-high sign pattern recurs in the guarded 2017–2024 era (historically contacted, not an independent replication). Activity means eligible-row count under the no-trade band, not volume, liquidity, or volatility. The dummy floors

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and 50-permutation within-day label-shuffle sentinel reduce class-imbalance and within-day-leakage explanations, but they have no power against a bid-ask-bounce alignment [29]. We therefore report the map as a diagnostic boundary, not model-attributable predictability.

The paper therefore claims a guarded evaluation procedure exercised on one near-null case. In this case, the procedure records estimand sensitivity, a weak guarded bar, multiplicity fragility, and a conditional sign that a headline-only readout would hide. Whether it also flags their absence on a genuine signal is the positive-control next step.

2 Related Work

Asset-pricing and equity-prediction studies apply machine-learning models to financial panels [11, 15]. Intraday limit-order-book work applies related classifiers to high-frequency streams [39]. We reuse this work for a narrower estimand: drop-neutral, 5-minute direction classification with a same-row stratified dummy floor. A macro-F1 margin here is classification evidence, not investment performance.

A primary methodological risk is validation contact. Leakage in machine-learning studies and decision-time backtest leakage show that adaptive validation can inflate apparent performance [22, 38]; reusable-holdout and preregistration work supplies validity-preserving guards [8, 20]. Chronological evaluation is the temporal guard [21]. Finance addresses data-snooping with backtest-overfitting probability, reality-check tests, and multiple-testing corrections [2, 3, 17, 26, 30, 33]. We combine these standard guards for this return-free macro-F1 estimand: fixed label and band, chronological day blocks with day-local windows, a same-row stratified dummy, and a validation-budget ledger. We exercise it on a single near-null case, not a known-signal control. Return-based benchmarks and investing-strategy studies score later economic stages [24, 38]; our readout audits the upstream classification gate.

The model families are intentionally ordinary. TCNs, decomposition linear models, and gradient-boosted trees are established sequence or tabular baselines [1, 14, 23, 37]. We use them to exercise the protocol; we make no architecture claim and no final-model selection.

Selective classification and calibration enter here only as an adverse diagnostic on whether confidence scores rank cases usefully [7, 12, 16, 32]. Brier-score decomposition separates calibration error from resolution [6, 27]. A second boundary is microstructural. Prior intraday-return studies document cross-sectional periodicity, time-varying short-horizon predictability, and seasonality [4, 18, 19]. That literature is a foil for our condition map: its predictability is tied to timing, higher-activity states, or seasonal rebalancing, not the calm slice where our edge concentrates. Bid-ask bounce can create row-level structure preserved by within-day permutation [25, 29]. Our controls cannot yet rule it out, so we frame the activity-tercile result as a conditional limitation.

3 Task and Evaluation Protocol

The protocol freezes the label, splits, baselines, readout metric, and decision criteria before the first model score. Any reported margin is therefore prespecified, not a post hoc choice. Split-boundary controls fix which rows can be scored, and a validation-budget ledger records the logged route-scoring contact behind the margin [22, 26]. We demonstrate it on a single near-null case with no known-signal positive control, so its power to separate signal from noise is asserted by construction, not shown here.

3.1 Task

We frame this as drop-neutral binary intraday direction classification on 5-minute bars, resampled from 1-minute data for five US large-cap equities (CSCO, JPM, KO, MSFT, WMT). Each prediction unit is one ticker and one completed 5-minute target bar, scored up or down over a fixed forward horizon. Any edge this procedure retains is classification evidence, not economic value.

3.2 Label

Each row's target is the sign of the 9-bar-ahead forward return. With θ the no-trade band in return units, a target row is labeled up when the forward return exceeds $+\theta$ and down when it falls below $-\theta$; a row inside the band is flagged invalid and dropped. We fix the band at 3.0 bps, which removes near-flat moves whose direction is dominated by noise [31, 35] and changes the label distribution relative to all bars [36]. It also defines the per-day activity proxy used in a later conditional diagnostic, a limitation rather than a result. Labels and windows are day-local and per-ticker: neither the horizon window nor a label crosses a trading-day boundary or a ticker. The remaining up/down rows are the only supervised rows; no-trade rows are not a third class and are not imputed.

3.3 Chronological splits

Chronological day blocks define the partitions, fixed by the Stage 00 freeze [21]. Train runs from 1998-01-02 to 2013-09-16 (end exclusive); official validation runs from 2013-09-16 to 2017-01-25 (end exclusive); the holdout is closed from 2017-01-25. No stage in the validation pipeline reads or scores a post-2017 bar; the run manifest records holdout contact as false. That seal is pipeline-scoped, not a guarantee the region is unseen. Earlier experiments outside that pipeline had already touched the post-2017 region. The guarded walk-forward readout there (Section 7) is therefore historically contacted, not a clean test. Because every window and horizon stays inside one trading day, the last train and first evaluation target windows cannot straddle the cut or share rows. The template deliberately substitutes this within-day locality for purge and embargo, and we apply neither. Cross-day serial dependence therefore stays intact, a limitation rather than a solved problem. Preprocessing that learns statistics is fit on train rows only and applied unchanged to validation. The feature window $w = 20$ is likewise fixed by a train-inner feature and window search (Section 5), not as a validation-tuned readout parameter.

3.4 Baselines

Every model score is reported against same-row trivial baselines on the identical target rows within each readout domain. The validation criteria use a stratified dummy, which samples from the empirical train label prior, and a majority baseline, which always predicts the most frequent train label. The stratified dummy fixes a same-row label-prior macro-F1 floor for the drop-neutral target. A margin over it means "above train-prior guessing" on the same

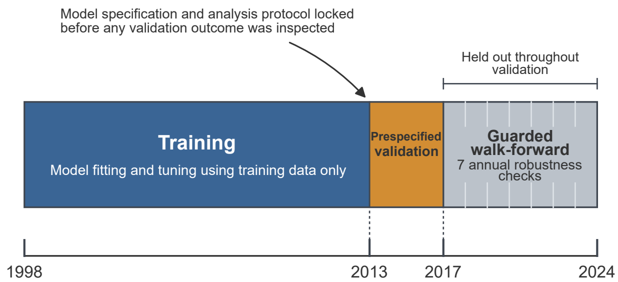

Figure 1: Protocol timeline and freeze ordering for the drop-neutral up/down direction task. Chronological day-blocks define training (1998-01-02–2013-09-16), one official validation readout (2013-09-16–2017-01-25), and a guarded post-2017 era. Model specification and analysis protocol are locked before validation; validation is prespecified, not tuning or model selection, and no final model is chosen. The post-2017 region was untouched during validation but had earlier out-of-pipeline contact, so the guarded walk-forward checks are historically contacted, not future-blind or independent hold-out tests.

rows; it does not by itself rule out serial structure or microstructure effects.

3.5 Readout metric and predeclared criteria

The model-versus-dummy margin is measured in macro-F1 on the drop-neutral up/down target. The decision criteria were predeclared in the dated protocol before any readout [20] and are evaluated on the predeclared TCN primary. On the validation split it passes when its macro-F1 delta is positive over both the stratified dummy and the majority baseline, and at least three of the five tickers are positive (a floor, not the achieved count). These are deliberately weak floors: predeclaration only prevents moving the bar after the fact; it does not make the bar discriminating. The guarded walk-forward readout carries its own separate, also weak, predeclared stability bar. Neither readout is a model-selection event, and no final model is selected.

3.6 Validation-budget ledger

Each readout contact with the official validation split is counted and tier-tagged in a validation-budget ledger, and every such contact carries `for_selection` set to false. This makes logged route contacts traceable within the route and bounds disclosed reuse of the official validation split; it is not an external timestamp or a reconstruction of earlier exploratory looks. The ledger later reports 2 official-validation seed-candidate scoring events and 56 guarded scoring events. The paper also keeps three evidence domains separate, each reported only within its own domain. These are the official validation readout over 2 seeds, the train-inner control comparisons, and the guarded historically-contacted walk-forward readout, also scored under the two prespecified seeds. Every reported number names its domain, and no sentence mixes them.

Figure 1 summarizes the timeline and the freeze ordering.

4 Models

The fixed roster only exercises the evaluation procedure; only the TCN is primary. We score exactly four families: a temporal convolutional network, a decomposition-linear model, a decomposition-convolutional hybrid, and gradient-boosted trees [1, 14, 23, 37]. None is a known-signal positive control. The roster can therefore show that the procedure does not manufacture an edge on a near-null case. It cannot show that it would catch a real one.

The predeclared primary is a temporal convolutional network [1] in a `tcn_tiny` configuration: two residual blocks of 16 channels each, kernel size 2, and learning rate 10^{-3}. Standard DLinear (Std-DLinear) splits each input window into trend and residual components, each fed to its own linear head before they are summed [37]. The hybrid pairs a multi-scale DLinear branch (MS-DLinear) with a small residual TCN branch (jointly, MS-DLinear+TCN); it is an alternative-architecture roster family, not a contributed model. The hybrid's MS-DLinear branch joins multiple decomposition kernels as in series-decomposition forecasters [34, 40]. The fourth family is LightGBM [23], a gradient-boosted tree run as a tabular baseline on windowed features [14].

The roster and the primary-versus-fallback rule are predeclared, and the train-inner search runs entirely before any validation or guarded readout. That search instantiates the predeclared TCN configuration above and reserves LightGBM as the primary-slot fallback for mechanical or reconstruction failures. None occurred, so the fallback never fires. Its output is frozen ahead of both readouts, and neither the validation nor the guarded readout selects a final model. On the guarded row-pooled mean some roster alternatives show numerically higher deltas than the predeclared TCN primary, but only post hoc and not like-for-like; the TCN is the lone pre-committed frozen primary. No recurrent model enters the roster. The last-step multilayer perceptron (`last_step_mlp`) appears only as a train-inner control, not a scored family. The four families share the data, seeds, and frozen designation that the next section fixes.

5 Experimental Setup

We fix the data and the train-inner search settings the protocol section defers here; Table 1 lists the frozen statistics. The study uses five large-cap US equities (CSCO, JPM, KO, MSFT, WMT) as 5-minute bars resampled from 1-minute data. The chronological split places 736,685 labeled target rows in the train interval (1998-01-02 through 2013-09-16, end exclusive) and 151,064 in the official validation interval (2013-09-16 through 2017-01-25, end exclusive). Per-ticker validation counts span ≈29.9k–30.8k. Labels use a 9-bar horizon and a 3.0 bps no-trade band. The input is a 20-bar window over the `price_volume_time` feature set, the output of the train-inner feature and window search; `price_action_core` is the registered fallback feature set. This step shortlists the feature set and window only; family designation is the separate train-inner step below.

This separate train-inner step fixes the predeclared primary and fallback designation and one profile per family. The frozen feature-window and train-inner configs, handoff, and readout records bind the grid, search-space files, inner folds, selected profiles, refit rows, early-stopping source, and resolved device. The bounded-profiles

Table 1: Frozen dataset statistics. Counts are eligible binary target rows after the no-trade band invalidates near-flat rows, over end-exclusive chronological day blocks. The holdout from 2017-01-25 received zero validation-phase contact (`holdout_test_contact`=false).

Property	Value
Tickers	CSCO, JPM, KO, MSFT, WMT (5)
Frequency	5-min bars (from 1-min)
Train interval	1998-01-02 – 2013-09-16 (excl.)
Validation interval	2013-09-16 – 2017-01-25 (excl.)
Train rows	736,685
Validation rows	151,064 (≈29.9k–30.8k/ticker)
Label horizon	9 bars
No-trade band	3.0 bps
Window w	20
Primary feature set	`price_volume_time`
Fallback feature set	`price_action_core`

search is a screening-grade train-inner selection, not an unconstrained or validation-tuned hyperparameter search. The official validation readout only characterizes this frozen specification and is not a model-selection event. The route manifest sets no_final_model_selected=true. We then report the official validation outcome in macro-F1 over n=2 seeds (101 and 202), refit on the recorded CUDA device; two seeds are too few to quantify seed-to-seed variance.

6 Validation Results

This section reports the official validation readout. Unless a sentence names another evidence domain, every number below is measured on the frozen validation split with the predeclared TCN primary, scored over $n = 2$ seeds. This is a two-seed readout, too small to estimate seed-to-seed uncertainty beyond the descriptive spread shown below. The readout is not a model-selection event: the route declares no final model and characterises that candidate rather than choosing among families.

The primary candidate meets the predeclared criteria. The predeclared TCN primary reaches macro-F1 of 0.5170 ± 0.0009 across 2 seeds, where the ± value is only the standard deviation (Table 2). It exceeds the same-row stratified dummy by 1.69 pp in the seed mean, with the smallest per-seed margin at 1.63 pp. It also exceeds the majority-class baseline by 18.8 pp. The per-ticker delta is a positive cross-seed point estimate in all five tickers. These three predeclared conditions are met. That is a statement of fact about a fixed bar, not a demonstration of effectiveness: the margin is a thin validation-domain edge above a near-even floor.

The floor sets the scale. The dummy baselines fix what "thin" means. The same-row stratified floor is roughly 0.499–0.501 in macro-F1, close to the even-class value, while the majority-class baseline collapses to 0.329 by abandoning one class entirely. The 18.8 pp gap over majority therefore reflects the stratified dummy's two-sided coverage, not a strong directional signal; the weak-signal claim rests on the 1.69 pp gap over the stratified dummy.

A train-inner control spread, not a family comparison. A separate evidence domain puts that margin in context. Train-inner control comparisons across four control rows (two of them last-step degenerate controls) show a 0.66 pp spread in macro-F1, against the separate 1.69 pp official-validation margin reported above. These four rows are train-inner diagnostic controls, not the guarded roster's full model families. The 1.69 pp margin and the 0.66 pp spread live in different evidence domains and are not an apples-to-apples pair. Read only as scale-setting, the comparison supports the paper's evaluation-discipline framing rather than a model claim; it does not rank model families on the validation split.

The primary is a TCN; the fallback never fires. The predeclared primary is the TCN configuration on the `price_volume_time` feature set with window $w = 20$. The LightGBM fallback is not activated: it is reserved for mechanical or reconstruction failures, none of which occurred, so no fallback path enters the readout.

The per-ticker edge is positive but uneven. The per-ticker deltas are all positive as cross-seed point estimates, yet they spread widely (Figure 2, Panel A). Across seeds, CSCO carries the largest cross-seed mean delta at +2.21 pp and JPM the smallest at +1.00 pp. All per-ticker intervals are descriptive rather than formal inference; the JPM block-bootstrap interval crosses zero, so the all-positive per-ticker count should not be read as an inferential claim. The pooled 1.69 pp margin is thus an average over a heterogeneous panel, not a uniform per-ticker effect, which is one reason the diagnostics in Section 8 interrogate where the same-row edge concentrates and inverts below the random prior.

How the bands and pooled margins are computed. The capped whiskers in Figure 2 are unadjusted 95% per-trading-day block-bootstrap intervals conditional on fixed seeds. Whole-day resampling handles within-day autocorrelation but not between-day regime clustering or cross-ticker common-factor dependence. Per-group target-row counts (roughly 30k in Panel A validation tickers and up to 55k in Panel B guarded periods) are therefore not independent samples, and the bands reflect scored-row sampling only. The first row in each panel is row-pooled, not averaged: per seed, pool target rows within the domain, recompute candidate and stratified-dummy macro-F1, take the delta, then average the two seed deltas.

7 Guarded Walk-Forward Readout

What the guarded readout is. To show what the procedure reports past the validation boundary, we run a guarded walk-forward readout over seven consecutive 12-month periods on the post-2017 segment. Per the seal of Section 3, that segment saw no validation-phase contact but was already touched by earlier exploratory work outside the pipeline. It is therefore guarded, not a future-blind test, and makes no clean-test, out-of-sample-proof, or final-model claim. The other roster rows are comparison evidence, never ranked or selected.

The bar is a weak floor, not the observed count. The predeclared TCN primary met the predeclared guarded stability criteria. The criteria are an intentionally weak floor: at least two of seven periods positive in macro-F1 over the same-row stratified dummy, and a

Table 2: Frozen validation split, predeclared TCN primary: seed means over $n = 2$ seeds (mean ± std where shown). Deltas are pp macro-F1 changes against named same-row baseline.

Quantity	Value
TCN primary macro-F1	0.5170 ± 0.0009
Δ vs. stratified dummy (seed mean)	+1.69 pp
Δ vs. stratified dummy (min seed)	+1.63 pp
Δ vs. majority class	+18.8 pp
Stratified dummy macro-F1	≈ 0.499–0.501
Majority-class macro-F1	0.329
Per-ticker Δ vs. stratified dummy (cross-seed mean)	
CSCO (max)	+2.21 pp
JPM (min)	+1.00 pp
Positive-ticker count	5/5

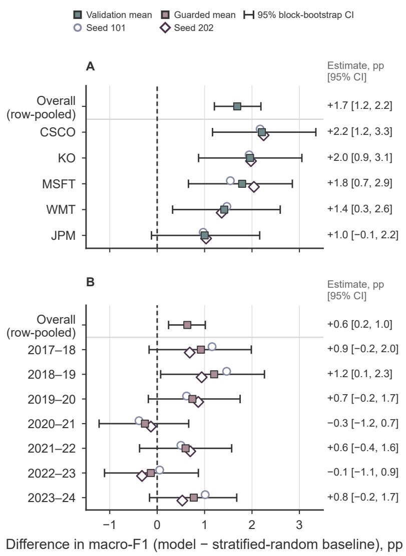

Figure 2: Descriptive macro-F1 deltas for the predeclared TCN primary against same-row stratified dummy; positive values favor the model. Panel A is prespecified validation by ticker; Panel B is the guarded, historically-contacted 12-month walk-forward. The JPM interval and six of seven guarded intervals include zero. Panel B is not a clean test, independent holdout, or model-selection ranking.

positive pooled delta. Under an independent fair-coin null, the two-of-seven clause alone passes about 94% of the time, so the bar only prevents post hoc relaxation. The observed five-of-seven positive periods (Figure 2, Panel B) are descriptive context, not certification and not the bar. Because the edge is near-null, we read the guarded readout as one worked instance of the procedure, not evidence of a durable edge. The procedure still reports what a headline-only readout would omit: a weak floor, a near-even multiplicity-discount readout, and a conditional sign.

Behind that pass, the period-level deltas are uneven. The readout comprises 56 guarded scoring events (seven periods, four roster rows, 2 seeds); the decision is judged on the primary's rows alone. The predeclared primary's 2020–21 and 2022–23 period means fall in the COVID and bear-market windows, and both are negative. The 2022–23 period also reverses sign across the two seeds, a dispersion 2 seeds are too few to quantify. The per-period readout is therefore not even seed-stable there. The addenda only re-aggregate frozen per-period prediction dumps; the fits are the predeclared V2.1 walk-forward refits that produced those dumps.

The headline delta is estimand-sensitive. The headline delta moves with the aggregation choice, so we report both units. The binding quantity is the row-pooled delta over the protocol row-union, +0.636 pp in macro-F1 over the stratified dummy; the equal-weight-per-period companion is +0.550 pp. Because the bar tests only the sign of a positive pooled delta, it is insensitive to this aggregation choice, and the headline magnitude is small under either unit.

Repeated comparison across the roster could also inflate this small edge, so we add a multiplicity discount. On the binding row-pooled estimand, we apply an odd-block CSCV/PBO adaptation (n_{trials}=4, n_{blocks}=7), not the canonical symmetric form. It yields a descriptive PBO readout of 0.514. That value is near a coin flip, too coarse to rule overfitting in or out [2, 3]. Only the predeclared TCN-primary period-LCB is above zero, at +0.047 pp; the worst-family LCB (Std-DLinear) sits at −0.464 pp [17]. These are descriptive discounts, not significance tests. On the row-pooled mean the primary (+0.636 pp) is third of four: LightGBM is largest (+0.726 pp), the hybrid MS-DLinear+TCN second (+0.672 pp, with the most positive periods at 6/7), and Std-DLinear fourth (+0.513 pp, 4/7). Yet all three non-primary families keep a period-LCB below zero. The three non-primary rows are post-hoc within-family profile maxima while the primary is the pre-committed frozen profile. Neither the mean ranking nor the period-LCB ranking is therefore a like-for-like family comparison. The primary's lone positive LCB is consistent with its single pre-committed profile, not a strongest-family claim, and no final model is selected.

Expanding-window training shares rows across periods, and the universe is a fixed five-ticker survivor set. The leave-one-out check is therefore largely mechanical here. We report it only as a descriptive sensitivity, not an independence or significance claim: the row-pooled delta falls no lower than +0.538 pp by period and +0.514 pp by ticker. The validation-budget ledger keeps the contact asymmetry explicit, at 2 official-validation scoring events against 56 guarded events, all with `for_selection=false`. The predeclared primary thus keeps the sign of its pooled edge under these drop-one perturbations, but only inside a guarded readout. The diagnostics section then treats the activity-tercile pattern as a diagnostic boundary. The same-row edge is largest in the calmest bars and drops below the balanced random prior in the most active ones. An open microstructure confound, however, prevents attributing that pattern to the model.

8 Diagnostics and Robustness

This section reports two limits on the readouts. The confidence-based acceptance signals of a near-base-rate classifier (calibration, selective prediction) are largely illusory, and the thin edge inverts across activity terciles.

Calibration error is small, but resolution is near zero. On the frozen validation split, the pooled probability output has an equal-mass ten-bin expected calibration error of approximately 0.010. The Brier score is 0.2496, but its vector decomposition is dominated by the uncertainty term (0.2499), leaving a resolution component of about 4.7×10^{-4} [6, 27]. A small calibration error therefore does not imply that the confidence scores rank cases usefully [16].

Selective prediction barely beats random ranking. On the same validation split, the accuracy-based risk–coverage curve (Figure 3) carries no cost model and marks no operating point. Its area (AURC) is approximately 0.470 against an oracle value near 0.140, an excess area (*e*-AURC) of about 0.330 in the seed mean. Read as descriptive within-seed, day-level summaries, not seed-level estimates over 2 seeds, the bands make the gain look thin. Against the random-ranking reference the predeclared score lowers AURC only marginally (ΔAURC $= -0.010$; 95% trading-day cluster-bootstrap CI $[-0.014, -0.006]$), about 3% of the gap between random ranking and a perfect-ranking oracle (gap-closed 95% CI $[1.9\%, 4.0\%]$). The gain shrinks further at high coverage: the mean random-minus-score selective-risk difference over the 80–100%, 90–100%, and 95–100% bands is only 0.25, 0.14, and 0.07 percentage points. The companion AUGRC is 0.237 [9, 12, 13, 32]. To the extent low-confidence rows fall in the high-activity tercile, abstention removes the classification rows where the below-random high-activity limitation is most visible, and so compounds it [7].

Architectural-component ablation moves little (train-inner domain). One train-inner control precedes the conditional split. Here `tcn_only` is the primary's TCN and `dlinear_only` is the hybrid's multi-scale DLinear branch (MS-DLinear) with its residual TCN removed. Each control row's gap to the reference stays within ± 0.7 pp of macro-F1 (`tcn_only` +0.03 pp, `dlinear_only` −0.14 pp, `last_step_lightgbm_control` −0.12 pp, `last_step_mlp` −0.63 pp; the latter two are last-step degenerate controls). The edge does not turn on the control-row choice, and this control is never merged with the validation or guarded readouts.

The edge concentrates in calm bars. We slice the same-row macro-F1 delta by activity terciles—the per-(ticker, trading-day) eligible-row-count proxy under the 3.0 bps no-trade band, not volume, liquidity, or volatility—rank-defined within ticker, so there is no single global cut point. On the validation split the delta runs +5.43 pp, +1.91 pp, and −1.54 pp from low to high activity tercile (Table 3). In the high tercile the macro-F1 (≈ 0.483) sits below the even-class prior (≈ 0.498) and the same-row stratified-dummy floor (≈ 0.500). The same direction recurs in the guarded 2017–2024 era: +4.08 pp, +0.56 pp, and −2.10 pp, with a high-tercile macro-F1 of 0.480 (Figure 4), the only axis we could re-measure on the guarded segment. This is a historically-contacted re-aggregation, not an independent replication. The figure's bars are across-seed means with open circle/diamond seed markers; its grey sleeves are per-trading-day block MDEs from official validation or measure-only re-aggregation of frozen guarded dumps, not standard errors or seed-level uncertainty. The signal is thus absent where the classifier faces the hardest slice, the most active bars.

Base-rate and within-day artifacts are excluded, but a microstructure artifact is not. The one alternative our controls cannot exclude is microstructure: the calm-bar concentration cannot presently be separated from a bid-ask-bounce alignment, and the sentinel has no power to do so. The bounce is a true row-level feature–label coupling that a within-day permutation preserves [25, 29], and the discriminating control needs raw five-minute bars the dump does not carry. The near-constant per-tercile dummy floor (≈ 0.500) rules out a class-imbalance artifact, while the up-rate shows only a small monotone gradient (0.523, 0.515, 0.502 from low to high, about 2.1 pp). A fifty-permutation within-day label-shuffle sentinel rules out within-day leakage. The observed delta exceeds the per-tercile shuffle maximum in every tercile, including the high tercile. There it exceeds a more negative shuffle null while staying below random. So the below-random value is itself not a permutation artifact. Because this confound remains open, we report the map as a diagnostic boundary on this estimand, not as model-attributable conditional predictability. If model-attributable it would run against prior intraday-return work tying short-horizon predictability to intraday timing, higher-activity states, or seasonal rebalancing [4, 18, 19]. A recent regime study does report forecastability in low-financial-uncertainty periods [10], a regime-level parallel, not one at our resolution. We carry the deferred microstructure control as a limitation (Section 9).

Table 3: Activity-tercile condition map. Validation uses 2 seeds and guarded is historically contacted. The high tercile falls below the balanced random prior in both domains; block-bootstrap checks are descriptive. Seed-mean rows (low/medium/high): 44,144/51,572/55,348 validation and 95,310/111,178/119,479 guarded.

Activity tercile	Validation Δ	Guarded Δ
Low activity	+5.43 pp	+4.08 pp
Medium activity	+1.91 pp	+0.56 pp
High activity	−1.54 pp	−2.10 pp
High-tercile macro-F1	≈ 0.483	0.480
Balanced prior	≈ 0.498–0.500	

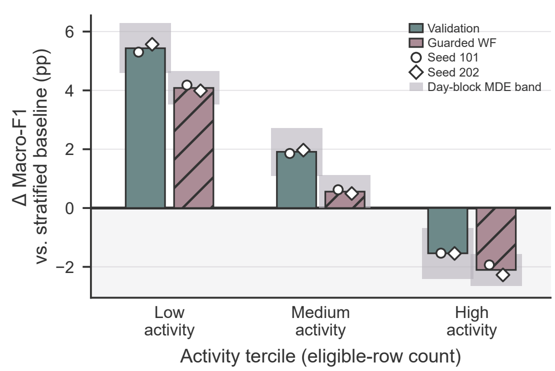

Figure 4: Activity-tercile condition map. Paired diverging bars compare validation and guarded macro-F1 deltas over the same-row stratified dummy; positive values favor the model. Activity is an eligible-row-count proxy under the no-trade band, not volume, liquidity, or volatility. The high-activity inversion is a limitation; guarded results are historically contacted, not a clean test, and grey sleeves are not confidence intervals.

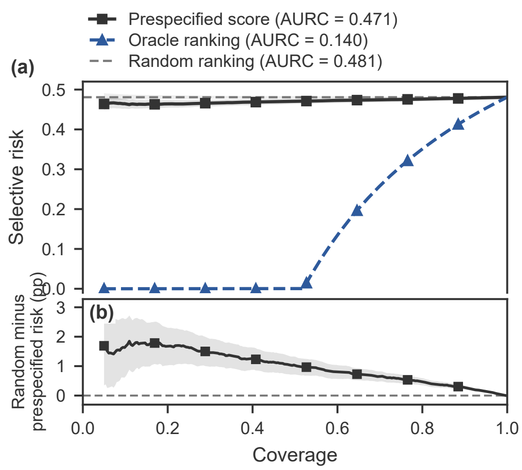

Figure 3: Accuracy-based selective-risk–coverage diagnostic; lower is better. The curve barely separates from random ranking, indicating weak confidence ordering rather than a usable abstention rule. There is no chosen operating point; with 2 training seeds, the plot establishes no statistical significance, calibration, or deployable abstention policy.

9 Conclusion and Limitations

The contribution is three-part. First, a strict pre-registered evaluation procedure whose pre-score freeze, same-row dummy floors, and budget ledger keep the readout auditable regardless of outcome. Second, under that procedure, a small margin for the predeclared frozen TCN primary over its same-row baselines, consistent in pooled sign but uneven across tickers and regimes—an evaluation finding, not an architecture claim. Third, feature and regime diagnostics (activity-tercile, calibration, selective-prediction) that bound and interpret that margin. We name the magnitudes to be scrutinized, not as effectiveness. A 1.69 pp validation margin and a 0.636 pp guarded row-pooled delta over the same-row dummy are thin edges above a near-even floor. The backtest-overfitting probability is a near-coin 0.514. The validation margin even flips sign across activity: from 5.43 pp on the calmest bars to 1.54 pp below the dummy on the most active—a conditional sign a headline-only readout would omit.

That finding stays narrow. The procedure is exercised on one near-null case with no known-signal positive control, so its sensitivity to a genuine signal is untested. The official validation readout rests on 2 seeds, thin for seed-to-seed variance [5, 28]. The Student-t lower bound shipped with the decision record is provenance, not a decision criterion. The scope is narrow: a single market and five large-cap survivors of the 1998–2024 span. Any positive readout is therefore entangled with survivor selection and does not extend to a broader panel.

The other limitations concern construct and claim scope. The no-trade band (3.0 bps) and the label horizon (nine bars) are pre-registered and frozen, with no threshold or horizon sensitivity scan. macro-F1 is a classification readout, not economic value. With no execution model or transaction-cost accounting, no margin here (validation, train-inner control, or guarded) is a trading claim. The guarded, historically-contacted walk-forward is not a clean test, and no final model is selected.

The guarded family margins are fragile: under a descriptive multiplicity discount on the binding row-pooled estimand, the discounted per-period lower bound falls below zero for all but the predeclared primary. This is not a like-for-like family ranking, since the non-primary rows are post-hoc maxima and the primary a frozen profile. The descriptive PBO readout is too coarse to rule overfitting in or out. The calm-bar concentration of Section 8 carries one open risk. The closing price close$[t]$ enters both the most-recent-return feature and the forward-label denominator. A bid-ask bounce is an alternative this classification-only evaluation cannot quantify, and it would be largest relative to the small returns of the calmest bars [25, 29]. The guarded label-shuffle sentinel rules out within-day leakage and day-base-rate artifacts, but has no power against the bounce, a true row-level coupling a within-day permutation preserves. The discriminating control compares a half-spread proxy against the 3.0 bps band, and needs raw five-minute bars the dump does not carry, so we defer it.

Confirming that the same procedure also flags their absence on a genuine signal is the defined next step, a known-signal positive control. Until then the outcome is a weak, conditional readout, not a model or a trading system.

Generative AI Usage Statement

Generative AI tools were used to assist with literature search, drafting and translating prose from the authors' notes, editing for style, generating figure-plotting code, and reviewing claim consistency, citations, provenance, and final-form issues. All experimental design, code execution, results, and claims were produced and verified by the authors. All AI-assisted text was reviewed, fact-checked against the project's experiment artifacts, and revised by the authors, who take full responsibility for the content.

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